

Taking Input Correctly

20 Practice Problems — Avoiding Common Input Mistakes

Instructions: For each scenario below, write the code that reads the input correctly and produces the required output. Think about what a student commonly gets wrong for that kind of input (extra spaces, wrong separator, wrong data type, missing validation, etc.) and make sure your solution avoids that mistake. You may use any language you are comfortable with; the model solutions are shown in Python.

Q1. Two Integers on One Line

Read two integers given on the same line, separated by a single space, and print their sum.

Q2. Float With Stray Spaces

Read a single floating-point number that may have extra leading or trailing spaces (e.g. ' 3.14 ') and print it multiplied by 2.

Q3. N Followed by N Integers

Read an integer N on the first line, then N integers on the next line separated by spaces, and print their sum.

Q4. A Full Name (With Spaces)

Read a person's full name, which may contain spaces (e.g. 'Rohiy K Sharma'), on a single line, and print it in uppercase.

Q5. A Single Character

Read one character typed by the user and print 'Valid Letter' if it is an alphabet (A-Z or a-z), else print 'Invalid'.

Q6. Comma-Separated Integers

Read a line of comma-separated integers (e.g. '4,9,2,7') and print them as a list, then print their total.

Q7. Handling Non-Numeric Input Safely

Read a value that is supposed to be an integer, but the user might type letters by mistake (e.g. 'abc'). Print the doubled value if valid, or 'Invalid number, please enter digits only' if not — without the program crashing.

Q8. Unknown Number of Lines (Read Until End)

Read integers, one per line, until the input ends (EOF) — there is no count given up front. Print their sum at the end.

Q9. A 2D Matrix

Read two integers R and C on the first line (rows and columns), then R lines each containing C space-separated integers, and print the matrix back out row by row.

Q10. Yes/No Input, Case-Insensitive

Read a Yes/No answer from the user, which might be typed as 'yes', 'Yes', 'YES', 'y', or with extra spaces, and print 'Confirmed' or 'Declined' accordingly.

Q11. A Value That Could Be Int or Float

Read a number that might be entered as a whole number ('7') or a decimal ('7.5'), and print it correctly as an int or a float depending on what was typed, then print its square.

Q12. Counting Words With Irregular Spacing

Read a sentence where words might be separated by multiple spaces (e.g. 'hello world there'), and print the correct word count.

Q13. A Percentage Value Like '85%'

Read a value typed like '85%' (with the percent sign attached) and print it as a plain number (85.0) that can be used in calculations.

Q14. A Date in DD-MM-YYYY Format

Read a date typed as 'DD-MM-YYYY' (e.g. '05-09-2026') and print the day, month, and year separately as integers.

Q15. Reading Negative Numbers Correctly

Read a line containing several space-separated integers that may include negative numbers (e.g. '-5 3 -2 8') and print their sum, making sure the negative sign is not mistaken for a separator.

Q16. Treating a Blank Line as Zero

Read a line that is supposed to contain a number but might be left blank by the user (an empty line). If blank, treat the value as 0; otherwise convert it normally, then print double the value.

Q17. Numbers Separated by Inconsistent Whitespace

Read a line of integers that might be separated by single spaces, multiple spaces, or tabs (e.g. '3 7\t2 9'), and print their sum correctly regardless of the spacing used.

Q18. A Row of 0s and 1s as Booleans

Read a line of space-separated 0s and 1s (e.g. '1 0 0 1 1') and print the equivalent list of True/False values.

Q19. A Fraction Like '3/4'

Read a fraction typed as 'numerator/denominator' (e.g. '3/4'), validate that the denominator is not zero, and print the decimal value.

Q20. A Very Large Number as Text

Read a number that may be too large for a normal integer to represent safely in some languages (e.g. a 25-digit number), and print the sum of its digits by processing it as a string instead of converting it to a numeric type first.